

LindaJamii Project Proposal

Title Page

PROJECT TITLE: LindaJamii: A Community Safety and Information Platform

PRESENTED BY: STUDENT NAME: [Your Name] REG NO: [Your Registration Number]

SUPERVISED BY: [Supervisor's Name]

PROPOSAL SUBMITTED TO THE SCHOOL OF COMPUTING AND INFORMATION TECHNOLOGY IN PARTIAL FULFILLMENT FOR THE BACHELOR OF / DIPLOMA IN TECHNOLOGY IN COMPUTER TECHNOLOGY PROJECT OF THE TECHNICAL UNIVERSITY OF KENYA

Submission Date: May 14, 2026

Declaration Page

I, [Your Name], declare that this project proposal is my original work and has not been submitted to any other university or institution for a degree or diploma. All sources used have been acknowledged.

Signed: _____ Date: _____

Dedication Page (Optional)

This project is dedicated to the communities striving for safety and information, and to the continuous advancement of technology for social good.

Acknowledgement Page (Optional)

I would like to express my sincere gratitude to my supervisor, [Supervisor's Name], for their invaluable guidance and support throughout this project. I also thank the Technical University of Kenya for providing the resources and environment necessary for this research.

List of Abbreviations

- API: Application Programming Interface
- CRUD: Create, Read, Update, Delete
- ERD: Entity-Relationship Diagram
- HTML: HyperText Markup Language
- JWT: JSON Web Token
- PDF: Portable Document Format
- REST: Representational State Transfer
- SCIT: School of Computing and Information Technology
- SQS: Simple Queue Service
- STK: Sim Tool Kit
- UI: User Interface

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Abstract

This proposal outlines the development of LindaJamii, a comprehensive community safety and information platform designed to enhance communication, incident reporting, and access to vital information within local communities. The platform integrates a multi-language backend (C, Python, Java) with a modern web frontend, featuring real-time incident tracking, M-Pesa donation capabilities, and an InfoHub for weather and news updates. LindaJamii aims to address the challenges of timely information dissemination and community engagement in safety initiatives, providing a robust, scalable, and user-friendly solution. This document details the project’s objectives, methodology, system analysis, design, implementation plan, and expected outcomes, adhering to the guidelines set forth by the Technical University of Kenya, School of Computing and Information Technology.

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CHAPTER ONE: INTRODUCTION

1.1 Introduction

The rapid urbanization and increasing complexity of modern societies have brought forth significant challenges in maintaining community safety and ensuring timely access to critical information. Traditional methods of communication often fall short in providing immediate updates on incidents, weather changes, or geopolitical shifts that directly impact residents. LindaJamii, a Swahili term meaning “Protect Community,” is envisioned as a digital solution to bridge these gaps, fostering a more informed, responsive, and resilient community.

This project proposes the development of a full-stack web application that leverages a diverse technology stack—including C for low-level services, Python (Flask) for a robust REST API, Java (Spring Boot) for enterprise-grade backend services, and a modern HTML/CSS/JavaScript frontend. The platform will offer features such as incident reporting, community alerts, M-Pesa donation integration, and an InfoHub providing real-time weather forecasts and curated news on medical, geopolitical, and climatic changes. The goal is to create a centralized, accessible, and interactive hub for community safety and information exchange.

1.2 Background of the Study

1.2.1 Background of the Organization

LindaJamii is a conceptual platform developed as a community-driven initiative. It operates on the principle that an informed community is a safe community. The platform is designed to be adopted by local community organizations, non-governmental organizations (NGOs), or even local government bodies to empower citizens with tools for collective safety and awareness. Its growth is predicated on its ability to provide reliable, real-time information and facilitate effective communication channels among residents and relevant authorities.

1.2.2 Overview of Existing System

Currently, many communities rely on informal communication channels, social media groups, or traditional media for safety alerts and information. These systems often suffer from fragmentation, delayed information, lack of verification, and limited reach. For instance, incident reporting might be manual, leading to slow response times. Weather updates might be generic, not localized, and news feeds might lack specific relevance to community safety. There is a clear absence of a unified, interactive platform that consolidates these critical functions.

Problems/Disadvantages of Existing Systems:

- **Fragmentation:** Information is scattered across multiple, unintegrated platforms.
- **Delayed Information:** Critical updates may not reach residents in a timely manner.
- **Lack of Verification:** Information shared informally can be unverified, leading to misinformation.
- **Limited Reach:** Not all community members have access to or are part of existing digital groups.
- **Inefficient Incident Reporting:** Manual processes hinder rapid response and coordination.
- **Lack of Centralized Data:** No single source for comprehensive community-relevant data (incidents, weather, news).

1.2.3 Overview of the Proposed System

LindaJamii proposes a unified, full-stack web application that addresses the shortcomings of existing systems. It will provide a centralized platform for:

- **Incident Reporting:** Users can quickly report incidents, which are then processed and displayed.
- **Community Alerts:** Dissemination of urgent safety alerts and notifications.
- **Real-time Information:** An integrated InfoHub for localized weather forecasts and curated news (medical, geopolitical, climatic).
- **Donation Gateway:** Secure M-Pesa integration to facilitate community funding and support.
- **Multi-language Backend:** Robust and scalable services built with C, Python, and Java.
- **Modern Frontend:** An intuitive and responsive user interface for seamless interaction.

Benefits of the Proposed System:

- **Enhanced Communication:** Real-time information flow improves community awareness.
- **Faster Response Times:** Streamlined incident reporting enables quicker action.
- **Data-Driven Insights:** Centralized data can be analyzed for safety trends and resource allocation.
- **Community Empowerment:** Provides tools for active participation in safety initiatives.
- **Financial Sustainability:** M-Pesa integration allows for direct community support.
- **Scalability and Robustness:** Built with a diverse and modern technology stack for future growth.

1.3 Problem Statement

The absence of a cohesive and efficient digital platform for community safety and information dissemination in many regions leads to fragmented communication, delayed responses to critical incidents, and a general lack of coordinated efforts in addressing local challenges. Existing informal systems are often unreliable, lack verification mechanisms, and fail to provide comprehensive, real-time updates on diverse threats ranging from local incidents to broader medical, geopolitical, and climatic changes. This deficiency directly impacts community resilience, preparedness, and the overall well-being of residents, necessitating a robust, integrated, and accessible solution.

1.4 Objectives

1.4.1 Project Goal (Major Objective)

The overall goal of this project is to design, develop, and deploy LindaJamii, a comprehensive, full-stack community safety and information platform that effectively integrates multi-language backend services with a modern web frontend to provide real-time incident reporting, community alerts, M-Pesa donation capabilities, and an InfoHub for critical weather and news updates, thereby enhancing community resilience and preparedness.

1.4.2 General Objectives

- To establish a robust and scalable backend infrastructure capable of handling diverse data processing and API requests.
- To develop an intuitive and responsive web frontend that provides a seamless user experience for information access and interaction.
- To integrate external services for real-time data, such as weather forecasts and global news, relevant to community safety.
- To implement a secure and efficient payment gateway for community donations.
- To ensure the platform is maintainable, extensible, and adheres to modern software development best practices.

1.4.3 Specific Objectives

- To implement a C-based micro-service for high-performance utility functions (e.g., hashing, health checks).
- To develop a Python (Flask) REST API for managing incident reports, alerts, and community statistics, including asynchronous processing for incident handling.
- To build a Java (Spring Boot) REST API for managing neighbourhood registries and patrol scheduling, incorporating Spring Security for access control.
- To design and implement a modern HTML/CSS/JavaScript frontend with a 5-step user flow (Splash, Sign Up, Verify, Onboard, Dashboard) and dedicated components for incident reporting, donations, weather, and news.
- To create a comprehensive database schema for managing users, posts, comments, followers, likes, chats, messages, and notifications.
- To integrate the Safaricom Daraja API for M-Pesa STK Push donations.
- To integrate a Weather Forecast API (e.g., OpenWeatherMap) and a News API (e.g., NewsAPI) to provide relevant information.
- To document the system architecture, API specifications, database schema, queueing mechanisms, deployment strategies, and security measures.

1.5 Justification

The development of LindaJamii is justified by the pressing need for improved community safety and information access. The platform offers a unique blend of features that directly address the limitations of current systems, providing a centralized, reliable, and interactive solution. By integrating diverse technologies, LindaJamii demonstrates a robust and scalable approach to complex system design, offering significant educational value in software engineering and distributed systems. The M-Pesa integration taps into a widely used mobile money ecosystem, ensuring accessibility for donations and fostering financial support for community initiatives. Furthermore, the real-time InfoHub provides critical, actionable intelligence, making the community more resilient to various threats. This project is not only technically challenging but also socially impactful, aligning with the goals of leveraging technology for societal benefit.

1.6 Scope of the Study

The scope of this project encompasses the design, development, and initial deployment of the LindaJamii web platform. This includes the full-stack implementation of the frontend and backend services as described in the objectives. The study will cover the integration of external APIs (M-Pesa, Weather, News), the design of a relational database schema, and the documentation of architectural patterns, queueing mechanisms, and deployment strategies. The project will focus on core functionalities for

incident reporting, information dissemination, and community engagement. While the platform is designed for scalability, the initial deployment will target a specific community context for demonstration and testing purposes. The study will not delve into mobile native application development beyond the web frontend, nor will it cover advanced machine learning models for predictive analytics, though the architecture will allow for future extensions.

1.7 Limitations of the Proposed System

- **API Rate Limits:** Reliance on third-party APIs (M-Pesa, Weather, News) may introduce rate limiting constraints, affecting the frequency of data updates or transaction processing under heavy load.
- **Data Accuracy:** The accuracy of information from external APIs is dependent on the providers.
- **Security Scope:** While robust security measures (JWT, bcrypt) are planned, a full-scale penetration test and security audit are beyond the scope of this project.
- **Scalability Testing:** Comprehensive load testing and performance optimization for extreme user loads will be limited by available resources and time.
- **Geographical Coverage:** Initial weather and news data may be focused on specific regions, requiring further configuration for broader global coverage.

1.8 Project Risk and Mitigation

Risk	Mitigation Strategy	Owner	Due Date
1. Data Breach	Implement strong encryption and access controls.	IT Department	2024-06-30
2. System Downtime	Regular backups and disaster recovery planning.	IT Department	2024-07-15
3. Compliance Violations	Conduct regular audits and employee training.	Legal & Compliance	2024-08-01
4. Talent Shortage	Invest in employee development and recruitment.	HR Department	2024-09-01
5. Financial Instability	Diversify revenue streams and manage expenses.	Finance Department	2024-10-01

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